

Group02 - PA2 - Project Proposal

Project Information

- **Course:** Human - Computer Interaction
- **Topic:** Running Music Coach - Adaptive Music and Eyes-Free Interaction for Runners
- **Team:** Group02
- **Lecturer:** Le Khanh Duy
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Members

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1. Project Title

Running Music Coach: Adaptive Music and Eyes-Free Interaction for Runners

2. Background and Rationale

The user research shows a repeated contradiction in current running experiences:

- music is essential for motivation and rhythm;
- pace awareness is important, but visual checking is disruptive;
- current phone/watch interaction becomes unreliable when users are moving and sweating.

As a result, runners already create manual workarounds such as pre-building playlists, disabling voice coach, turning on Do Not Disturb, and avoiding interaction unless absolutely necessary. These patterns suggest an opportunity to design for running flow rather than adding more intrusive coaching.

3. Problems the Team Wants to Solve

Based on the research, the team decided to focus on three severe and recurring problems that are not sufficiently solved by current running apps.

Problem 1 - Music Flow and Motivation

Voice prompts, notification ducking, and system interruptions cut into music, reduce motivation, and break emotional rhythm during solo runs.

Problem 2 - Eyes-Free Pace Awareness

Runners want to know whether they are too fast, too slow, or overexerted without repeatedly looking at a screen.

Problem 3 - Reliable In-Run Interaction

Sweat, movement, lock screens, and small watch displays make pause/resume, stat checking, and music control difficult and error-prone.

4. Overall Concept of the Solution

The team proposes a **hybrid phone-and-watch running support system** called **Running Music Coach**.

Conceptually:

- the **phone app** manages running session logic, music adaptation, and notification policy;
- the **watch** acts as a low-attention feedback and quick-control surface when available;
- **earbuds** provide lightweight interaction and ambient cue delivery;
- the system helps users stay in flow without forcing repeated visual attention.

This is not just a standard running tracker. It is a running companion centered on **adaptive music**, **subtle feedback**, and **reduced interaction cost**.

5. Conceptual Solutions per Problem

For each major problem, the team generated 2 to 3 conceptual solutions that differ significantly from each other, then selected the most promising combination.

5.1 For Problem 1 - Music Flow and Motivation

Concept 1A - Focus Run Mode

A run-specific mode that suppresses or delays non-critical notifications while the run is active.

Strengths

- protects music continuity;
- easy for users to understand;
- works well across phone-first and watch-assisted users.

Limitations

- does not directly improve pace awareness;
- still depends on user-configured exception rules.

Concept 1B - Ambient Audio Cues

Replace spoken coaching with subtle sounds or embedded cues that blend into music rather than cutting it off.

Strengths

- preserves emotional flow better than voice prompts;
- reduces irritation from abrupt interruptions.

Limitations

- cue design must be subtle but still understandable;
- may be less accessible for some users than speech.

Concept 1C - End-of-Segment Coaching

Delay non-urgent coaching messages until the user finishes a segment or briefly slows down.

Strengths

- avoids interrupting high-focus moments;
- still delivers meaningful progress information.

Limitations

- delayed feedback may be less useful for immediate correction;
- requires good timing logic.

5.2 For Problem 2 - Eyes-Free Pace Awareness

Concept 2A - Haptic Pace Feedback

Use distinct vibration patterns to indicate "too fast," "too slow," or "in range."

Strengths

- truly eyes-free;
- suitable for runners who dislike voice coaching.

Limitations

- patterns must be easy to distinguish while fatigued;
- depends on watch or device vibration quality.

Concept 2B - Beat-to-Pace Matching

Use music tempo or background rhythm as the main mechanism to help runners maintain target cadence.

Strengths

- directly connects pacing with music-driven behavior;
- fits the observed role of music in motivation and rhythm.

Limitations

- may not fit all music preferences;
- more complex to implement well.

Concept 2C - Pace Zone Visual Lite

Use a simplified one-state visual indicator, such as color or icon, for quick glances instead of a dense dashboard.

Strengths

- easier to understand than multiple numeric metrics;
- good fallback for users who still want occasional visual checks.

Limitations

- still depends on a screen glance;
- weaker than fully eyes-free methods.

5.3 For Problem 3 - Reliable In-Run Interaction

Concept 3A - Run Lock Interface

A minimal run screen with only essential actions and large high-contrast targets.

Strengths

- lowers mis-tap risk;
- reduces UI complexity during movement.

Limitations

- still touch-based;
- may not fully solve hands-free needs.

Concept 3B - Earbud Controls Integration

Allow pause/resume, skip, and cue acknowledgment through earbud gestures.

Strengths

- reduces phone/watch handling;

- supports more natural in-motion interaction.

Limitations

- depends on earbud capability and gesture consistency;
- not every user owns compatible earbuds.

Concept 3C - Auto Pause Smart Logic

Detect short stops and automatically pause/resume without requiring manual action.

Strengths

- removes repetitive interaction entirely in common stop scenarios;
- especially useful at traffic lights or short pauses.

Limitations

- false positives or negatives can frustrate users;
- requires careful tuning.

6. Selected Proposal

After comparing alternatives, the team selected a combined solution:

Running Music Coach

The chosen concept combines:

- **Focus Run Mode** from Problem 1
- **Ambient Audio Cues** from Problem 1
- **Haptic Pace Feedback** and selective **Beat-to-Pace Matching** from Problem 2
- **Run Lock Interface** and **Earbud Controls Integration** from Problem 3

This combination best addresses the core breakdowns observed in research while remaining conceptually coherent.

7. Main Components of the Proposed Solution

7.1 Focus Run Mode

- suppresses non-urgent notifications during an active running session;
- keeps music playback stable and uninterrupted;
- allows urgent contacts or safety exceptions.

7.2 Adaptive Music Engine

- recommends or adapts tracks to better match target cadence or effort;

- avoids hard cuts and intrusive spoken overlays;
- helps reduce unintended pace spikes caused by very fast songs.

7.3 Ambient Audio and Haptic Feedback

- signals "too fast," "too slow," or "in range" through subtle cues;
- delivers feedback only when deviation lasts long enough to matter;
- prioritizes low-attention channels instead of voice interruption.

7.4 Run Lock Interface

- provides a minimal in-run screen with large controls;
- supports pause/resume and a small number of essential actions;
- reduces unlock-and-navigate steps during movement.

7.5 Earbud Quick Controls

- supports skip, pause/resume, or cue acknowledgement where hardware allows;
- reduces dependence on direct screen interaction.

8. Why This Proposal Fits the Research

- It addresses the motivation problem by protecting music continuity.
- It addresses the pace-awareness problem by shifting feedback away from constant visual checking.
- It addresses the interaction problem by minimizing mid-run touch effort.
- It builds on user workarounds already observed in the field instead of ignoring them.

9. Target Users

- **Primary:** young recreational runners aged 18-25
- **Key subgroups:** phone-first runners, watch-assisted runners, and music-driven solo runners

10. Scope

In Scope

- running-session setup and start;
- in-run feedback and interruption handling;
- quick control during movement;
- end-of-run summary and sync.

Out of Scope

- full training-plan management;

- competitive social platform features;
- advanced clinical diagnostics.

11. Expected Benefits

For users, the solution is expected to:

- improve motivation retention during solo runs;
- reduce flow-breaking audio interruptions;
- reduce the need for repeated screen glances;
- lower mis-tap frustration during movement;
- support a more eyes-free and music-centered running experience.

12. Success Criteria

- reduced frequency of manual screen checks;
- reduced complaints about music interruption;
- improved perceived reliability of in-run controls;
- positive user feedback on flow, comfort, and motivation.